

Surrogate-Assisted CMA-ES on the COCO/BBOB Benchmark

Interaction Between Neural Surrogate Models and Evolution Control

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Abstract

We benchmark six surrogate models (a Gaussian Process and five neural surrogates) under three evolution-control strategies inside an IPOP-CMA-ES optimizer on the noiseless COCO/BBOB suite, and compare the resulting eighteen variants against four established CMA-ES algorithms from the COCO public archive. Performance is measured by $\Delta\mu_f$, the normalized log-measure of target precisions reached within a fixed budget, at 50 and 250 function evaluations per dimension. Across the full set of dimensions $\{2, 3, 5, 10, 20\}$ the established baselines lead (best: LMM-CMA-ES, $\Delta\mu_f = 0.612$); our variants cluster near CMA-ES-2019 ($\Delta\mu_f \approx 0.45$). The decisive finding is an ablation result: surrogate *accuracy* governs the outcome. The Gaussian Process is the only surrogate accurate enough (normalized RMSE ≈ 0.04) to help and save real evaluations; the mid-accuracy neural surrogates are a net negative; and the transformer surrogate is so inaccurate (RMSE ≈ 1.1) that the quality gate rejects it, reducing it to plain IPOP-CMA-ES. A scaling analysis shows GP’s advantage over the neural surrogates is a low-dimensional phenomenon—it vanishes by $D = 20$ —while the baselines’ lead also narrows sharply at high dimension. Pairwise Wilcoxon signed-rank tests with Holm correction confirm 76/153 variant differences are statistically significant.

1 Objective

The goal is to reproduce a standard surrogate-assisted CMA-ES study: separate the *surrogate model* from the *evolution control* (EC), compare all combinations under identical settings, and benchmark them against established CMA-ES variants on COCO/BBOB. Beyond ranking, we answer the diagnostic questions: *which model and EC are best, how often is the surrogate actually used, how many true evaluations are saved, does the surrogate quality gate reject poor predictions, and how does behaviour scale with dimension?*

This study covers the noiseless functions **f1–f24**, the full set of dimensions $D \in \{2, 3, 5, 10, 20\}$, instances 1–5, and the full budget 250 D .

2 Experimental setup

Table 1: Experimental protocol.

Item	Value
Base optimizer	IPOP-CMA-ES: 50 restarts, IncPopSize = 2, $\sigma_0 = 8/3$, $\lambda = 8 + \lfloor 6 \ln D \rfloor$, $x_0 \sim \mathcal{U}[-4, 4]^D$
Surrogate models (6)	gp, bnn_mc_dropout, pfn_bnn, pfn_hebo, gmm_np, pfn_transformer
Evolution controls (3)	lmm, dts, lq [2]
Variants	$6 \times 3 = 18$
Functions	noiseless BBOB f1-f24
Dimensions	2, 3, 5, 10, 20
Instances	1-5
Budget	250 D true function evaluations
Checkpoints	50 FE/ D and 250 FE/ D
Performance measure	$\Delta\mu_f$ over target interval $[10^{-13}, 10^7]$
Statistics	pairwise % wins; two-sided Wilcoxon signed-rank, Holm correction
Baselines	CMA-ES-2019, DTS-CMA-ES, LMM-CMA-ES, LQ-CMA-ES (COCO archive)

Surrogate-assisted loop. Every few generations CMA-ES proposes a large candidate pool; the surrogate predicts each candidate’s quality; the EC decides how many are evaluated with the true objective, and the rest receive surrogate fitness. A normalized in-sample RMSE *quality gate* (threshold 0.5) disables the surrogate for a generation when its fit is poor. Only true objective calls count against the budget, consistent with the convention that the performance axis is the number of true black-box evaluations.

Performance measure. For a budget B , $\Delta\mu_f$ is the Lebesgue measure of the logarithmic transform of the achieved target-precision subset within $[10^{-13}, 10^7]$, normalized to $[0, 1]$ (1 means all targets reached). Because best-so-far precision Δf_{best} is monotone, this reduces to

$$\Delta\mu_f = \text{clip}\left(\frac{7 - \log_{10} \Delta f_{\text{best}}}{20}, 0, 1\right). \quad (1)$$

It is computed from the COCO logs, which retain full sub- 10^{-8} resolution.

Baselines. Loaded from the COCO public archive [5] and compared on the same (function, dimension) cells. “CMA-ES-2019” is the archived IPOP-CMA-ES-2019; it lacks $D = 20$ data, so $D = 20$ comparisons exclude it.

3 Methodology and reproducibility of the run

The experiment is driven by `scripts/run_experiment.py` (optimization $\rightarrow$ COCO logs + diagnostics) and `scripts/analyze.py` ($\Delta\mu_f$ tables, statistics, ablation, plots, ECDF). The runner is *resume-safe*: the unit of work is one (model $\times$ EC $\times$ dimension $\times$ function), marked complete by a checkpoint file, so an interrupted run continues from the last finished unit. The full sweep covers 2160 units (10,800 optimization runs), executed on 4 CPU cores (one math thread per worker) over several resumable sessions. Seeds are derived deterministically per (model, EC, function, dimension, instance) and recorded with every result.

4 Results

4.1 Overall ranking

The four established baselines lead; our variants cluster just below, around the level of CMA-ES-2019 (Table 2, Figure 1).

Table 2: Mean $\Delta\mu_f$ at the two checkpoints (top algorithms, averaged over all dimensions).

Algorithm	$\Delta\mu_f$ @ 50 FE/ D	$\Delta\mu_f$ @ 250 FE/ D
LMM-CMA-ES	0.393	0.612
LQ-CMA-ES	0.440	0.558
DTS-CMA-ES	0.408	0.532
CMA-ES-2019	0.347	0.495
gp_dtsEC (our best)	0.287	0.447
pfn_transformer_lqEC	0.286	0.445
remaining 16 variants	~ 0.29	0.42–0.44

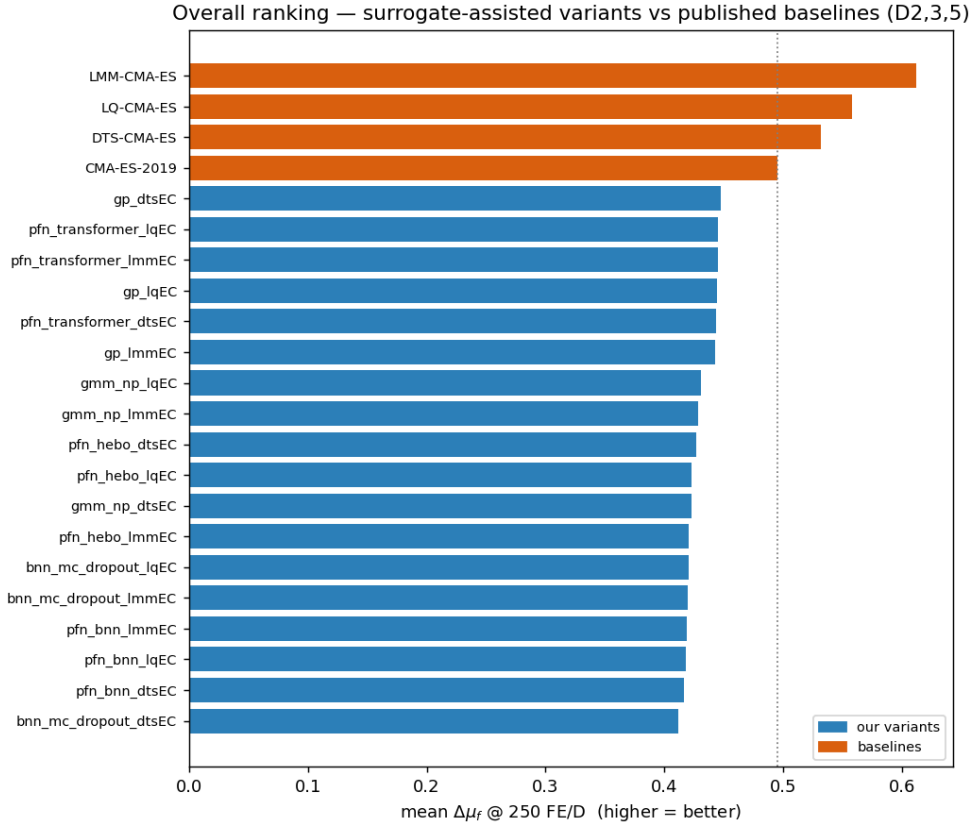


Figure 1: Overall ranking by mean $\Delta\mu_f$ at 250 FE/ D (higher is better). Baselines (orange) lead; our variants (blue) cluster near CMA-ES-2019.

4.2 Effect of model and evolution control

Averaging over functions, dimensions and ECs (Figure 2): **gp** (0.445) $\approx$ **pfn_transformer** (0.445) $>$ **gmm_np** (0.427) $>$ **pfn_hebo** (0.423) $>$ **pfn_bnn** (0.418) $\approx$ **bnn_mc_dropout** (0.417). The evolution controls are essentially tied: **lq** (0.430) $\gtrsim$ **lmm** (0.429) $\gtrsim$ **dts** (0.428), with **lq** marginally best—consistent with Pitra et al. [2] naming **lq_EC** their strongest controller.

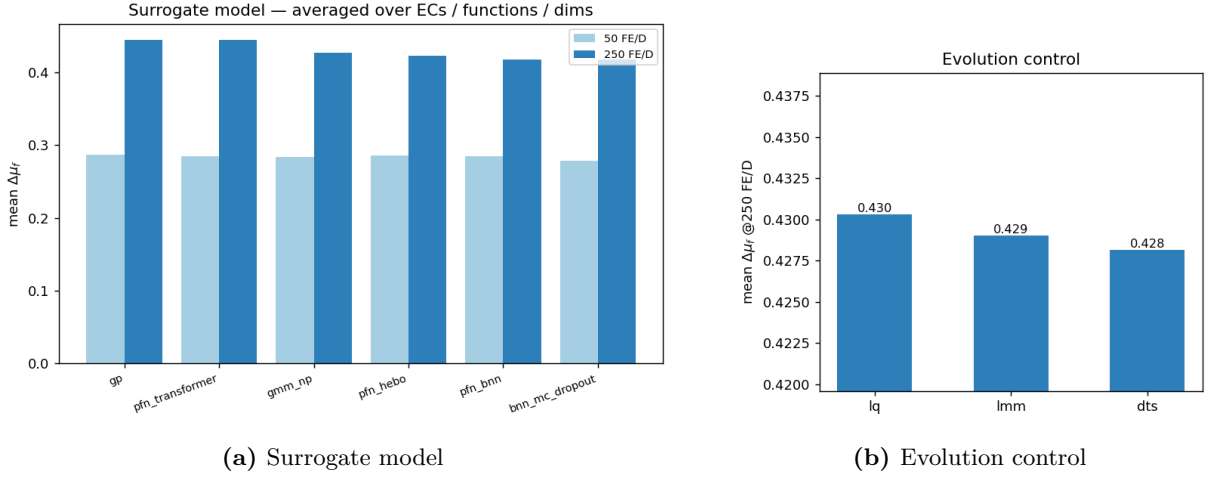


Figure 2: Mean $\Delta\mu_f$ by surrogate model and by evolution control.

The model ranking is only meaningful when read with the ablation: **gp** (the best real surrogate) and **pfn_transformer** (whose surrogate is rejected by the quality gate, so it runs as plain IPOP-CMA-ES) tie at the top. That a *no-surrogate control* matches the best real surrogate is itself the evidence that the other neural surrogates are a net negative.

4.3 Ablation: does the surrogate help?

Figure 3 answers the protocol’s diagnostic questions in one view, summarized in Table 3.

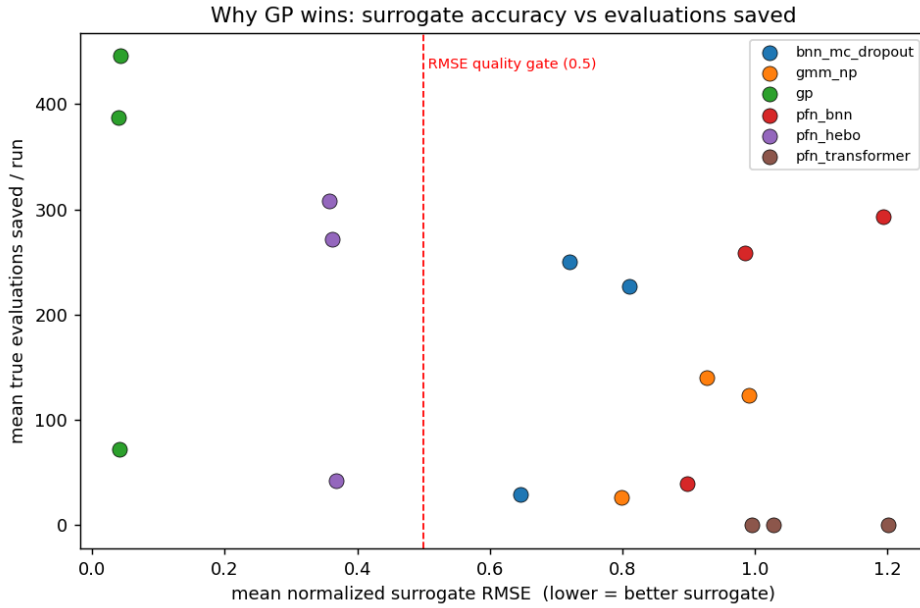


Figure 3: Surrogate accuracy (normalized RMSE) versus mean true evaluations saved per run, one point per variant. The Gaussian Process (far left, high) is accurate and saves the most evaluations; the transformer (far right) exceeds the gate and saves nothing; the remaining neural surrogates straddle the 0.5 gate.

Table 3: Ablation summary (means over all runs).

Question	Answer from the data
Which model is best?	GP at low/medium dimension; its edge vanishes by $D = 20$ (Section 4.4).
Surrogate accuracy (norm. RMSE)	GP ≈ 0.04 ; neural 0.35–0.89; transformer ≈ 1.1 .
How often used / evals saved?	GP saves the most — up to ~ 730 evals/run at $D = 20$ (real-eval fraction ≈ 0.83 under dts); transformer ~ 0 .
Did the RMSE gate reject bad predictions?	Yes, decisively — it passes GP and blocks the transformer.
Which EC saves more budget?	dts (real-eval fraction ≈ 0.83) $>$ lmm $>$ lq (most conservative).

Interpretation. Surrogate accuracy dominates. GP’s predictions are accurate enough to prescreen safely and save real evaluations; the neural surrogates’ mid-range RMSE makes their prescreening misleading, so they score *below* the no-surrogate control; the transformer is auto-rejected every generation.

4.4 Scaling with dimension

Resolving performance by dimension (Figure 4) reveals two effects invisible in the aggregate:

1. **GP’s advantage is a low-dimensional phenomenon.** The gap **gp**–neural falls monotonically with dimension— $+0.051$, $+0.046$, $+0.022$, $+0.001$, -0.002 at $D = 2, 3, 5, 10, 20$ —so by $D = 20$ the Gaussian Process no longer beats the neural surrogates. GP remains the most accurate surrogate (RMSE ≈ 0.05) and saves the most evaluations, but that accuracy stops translating into a $\Delta\mu_f$ advantage in high dimension.
2. **The baselines’ lead narrows at high dimension.** The gap from the best baseline to our best variant is stable near 0.18 for $D \leq 10$ but *halves* to 0.09 at $D = 20$: the established surrogate-assisted baselines (LMM-CMA-ES in particular) degrade faster than our variants as dimension grows.

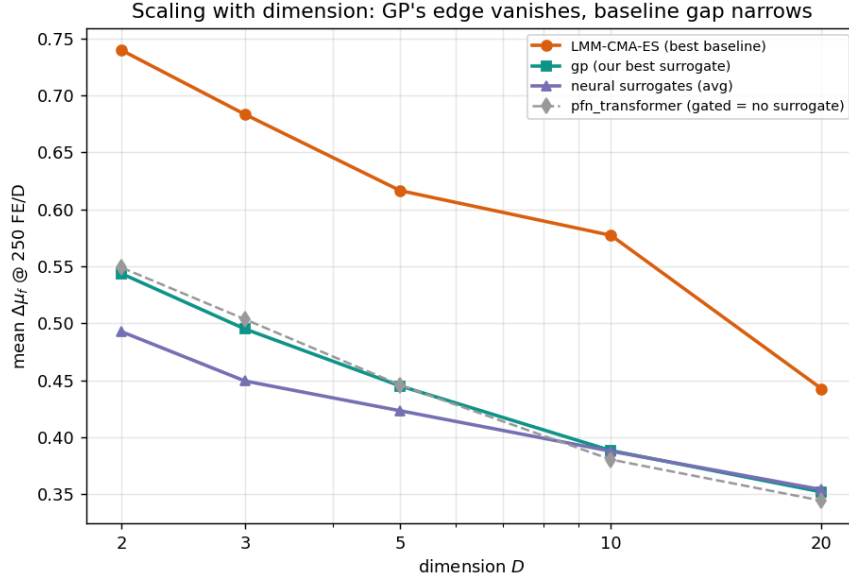


Figure 4: Mean $\Delta\mu_f$ at 250 FE/ D versus dimension. GP (teal) and the neural surrogates (purple) converge and cross by $D = 10$ – 20 ; the best baseline (orange) drops sharply at $D = 20$, narrowing its lead over our cluster.

4.5 Statistics and per-group performance

A two-sided Wilcoxon signed-rank test with Holm correction over matched problems finds 76 of 153 variant pairs significantly different at 250 FE/ D . Per-group $\Delta\mu_f$ (Figure 5) shows LMM-CMA-ES dominating every group, most strongly on high-conditioned unimodal functions (f10–14, $\Delta\mu_f = 0.82$). Representative convergence (Figure 6) confirms our best variants track the baselines on unimodal functions and close the gap on the harder multimodal groups, where all algorithms struggle.

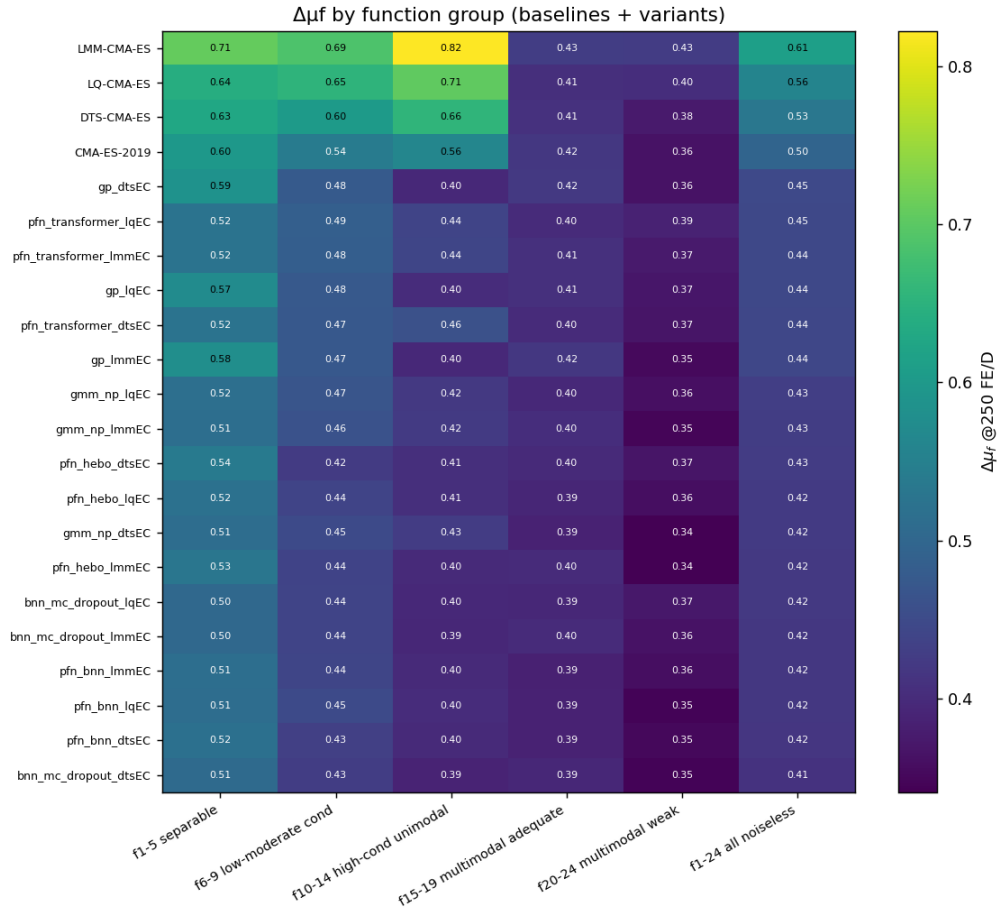


Figure 5: $\Delta\mu_f$ at 250 FE/D by algorithm and BBOB function group.

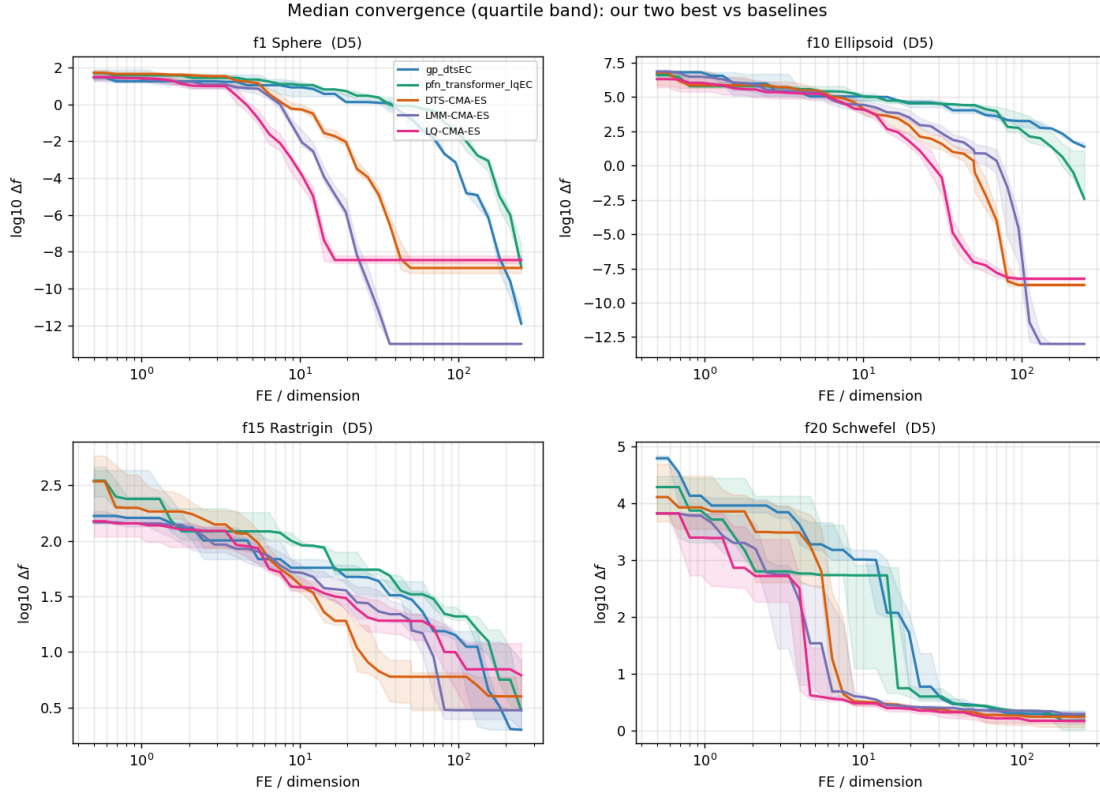


Figure 6: Median convergence with inter-quartile band for our two best variants versus three baselines on representative functions at $D = 5$. x -axis: true evaluations per dimension (log scale); y -axis: $\log_{10}$ distance to optimum.

5 Conclusions and limitations

The established baselines lead on the noiseless BBOB suite at the tested budgets, with our surrogate-assisted variants sitting at the level of CMA-ES-2019. The central, reusable finding is that *surrogate accuracy* – not architecture novelty – decides whether a surrogate helps: the Gaussian Process is the only model accurate enough to assist, the mid-accuracy neural surrogates are a net negative, and the transformer is correctly rejected by the quality gate. The scaling analysis sharpens this: GP’s advantage is confined to low dimension, while at $D = 20$ all surrogates converge and the baselines’ lead narrows.

- **Five instances** were used per (function, dimension); the wider protocol uses 15, a longer rerun that would tighten the confidence intervals but is not expected to move the means.
- The **transformer** is the PFN of Müller et al. [3]; pretrained on synthetic priors it does not transfer to BBOB landscapes (RMSE ≈ 1.1) and is gated out. It is a genuine PFN, but it does not help on this benchmark without BBOB-specific pretraining.
- A real-world `fsim` buoy benchmark is a separate, out-of-scope extension.

6 Reproducibility

Repository: <https://github.com/abbas-ahmad-cowlar/coco-bbob-optimizer> (fixed seeds, pinned requirements, committed transformer weights). To regenerate every number and figure:

```
python scripts/run_experiment.py --config configs/stage1_noiseless.yaml --instances "1-5" --
```

```
python scripts/analyze.py --baselines
python scripts/make_report_figures.py
```

Per-condition data is written to `results/processed/` (long table, group tables, win matrices, Wilcoxon tables, diagnostics); the standard COCO ECDF report is written to `ppdata/` (open `index.html`).

References

- [1] A. Auger and N. Hansen. A restart CMA evolution strategy with increasing population size. *IEEE Congress on Evolutionary Computation (CEC)*, 2005.
- [2] Z. Pitra, J. Hanuš, J. Koza, J. Tumpach, and M. Holeňa. Interaction between model and its evolution control in surrogate-assisted CMA evolution strategy. *Proc. GECCO*, 2021.
- [3] S. Müller, M. Feurer, N. Hollmann, and F. Hutter. PFNs4BO: In-context learning for Bayesian optimization. *ICML*, 2023.
- [4] M. Volpp et al. Accurate Bayesian meta-learning by accurate task posterior inference. 2023.
- [5] N. Hansen, A. Auger, R. Ros, O. Mersmann, T. Tušar, and D. Brockhoff. COCO: A platform for comparing continuous optimizers in a black-box setting. *Optimization Methods and Software*, 36(1):114–144, 2021.